ANALYSIS OF THE SHEAR ZONE MAP EXERCISE

The aim of this exercise is to describe the structures shown on the accompanying map and deduce the structural evolution of the area. Questions 1 to 3 take you through this and provide you with all the information needed to deduce the structural evolution. I could have asked you to write a 1-2 page (plus figures) evidence-based account of the structural evolution but instead have indicated below how you would go about doing this.

1. Complete the map. Draw on the fold axial traces and label them F_1 and F_2 as appropriate. A second deformation event affects the units in only a part of the map. Identify the features that provide evidence of this second deformation and then define the boundaries of the region that is affected by it. This region is a shear zone. From your observations compile a sequence of events.

2. Plot a stereogram to determine the geometry of the folding in those parts of the map that are affected ONLY by the first deformation event. Plot a second stereogram to determine the geometry of the folding inside the shear zone. Both stereograms need to be fully annotated and accompanied (on the same piece of tracing paper) by a simplified profile plane section of the folds.

3. Determine the attitude (strike/dip) of the shear zone. Determine the plunge/plunge direction of the slip vector. What is the sense of movement on the shear zone? Draw a fault plane section to determine the magnitude of the slip vector from the offset of the dykes and the orientation of the slip vector. Check your solution by drawing a fault plane section without using the slip vector, that is, by using the offset of the dyke and the offset of one of the early fold traces.

How you might lay out an evidence-based account of the structural evolution (using the indicated section headings)

This account should be in a logical order and should incorporate a full account of the geometry of the structures present on the map together with evidence for all your assertions about the timing and geometrical interrelationships.

- 1-2 sentence introduction (no heading required) which sets the scene for the reader by summarizing what is on the map. Use concise, informative and elegant phrasing.
- ***The geometry of the structures outside the D_2 shear zone.*** Describe the stratigraphy, the geometry of the folding (noting the structures associated with it), and the geometry of the dyke. Include as a figure supporting your text your stereogram of the folding outside the shear zone with simplified profile plane section.
- ***Observations that indicate D_2 is localized in a shear zone.*** Describe the structures that provide evidence that there is a shear zone crossing the map. Include as figures supporting your text your stereogram of the fold geometry inside the shear zone, and an annotated diagram illustrating the congruency of the structures formed during the shearing.
- ***The geometry of the D_2 shear zone.*** Explain how you can determine the attitude of the shear zone and the orientation of the slip vector. Include as a figure supporting your text the stereogram that you used to do this. Describe the geometry of the shear zone and slip vector, illustrating your text with the two fault plane sections that you have drawn.
- ***Sequence of events.*** Compile a bullet-pointed sequence of events. Each bullet should be accompanied by 1-2 sentences that provide evidence for the timing relationships and give key geometric numbers (*e.g.*, the strike/dip of the shear zone).